



EduSuite Pro

System Workflows & Lifecycle Architecture

A comprehensive blueprint documenting the core operations, student lifecycles, examination procedures, and administrative workflows.

DOCUMENT SCOPE:

- Academic Management Lifecycle
- Faculty Operations & HR Workflow
- Examination & Results Processing
- Student Management & Lifecycle Workflow
- Access Control & Admin/Super-Admin Architecture
- Training & Placement Cell Processes
- Library Operations & Member Management

METADATA:

Status: Approved Blueprint
Version: 2.0.0 (Enterprise)
Date: July 31, 2026
Target: Dev & QA Teams

Table of Contents

1. Academic Management Workflow	3
2. Faculty Operations & Management Workflow	4
3. Examination & Results Processing Engine	5
4. Student Management & Complete Lifecycle	6
5. Access Control & Admin/Super-Admin Architecture	8
6. Training & Placement (T&P) Cell Workflow	10
7. Library Administration & Operations Workflow	12

1. Academic Management Workflow

The Academic Management module drives the curriculum planning, program coordination, and academic scheduling of the institution. It bridges the core rules with operational entities like section mapping, timetable generation, and attendance validation before exams.

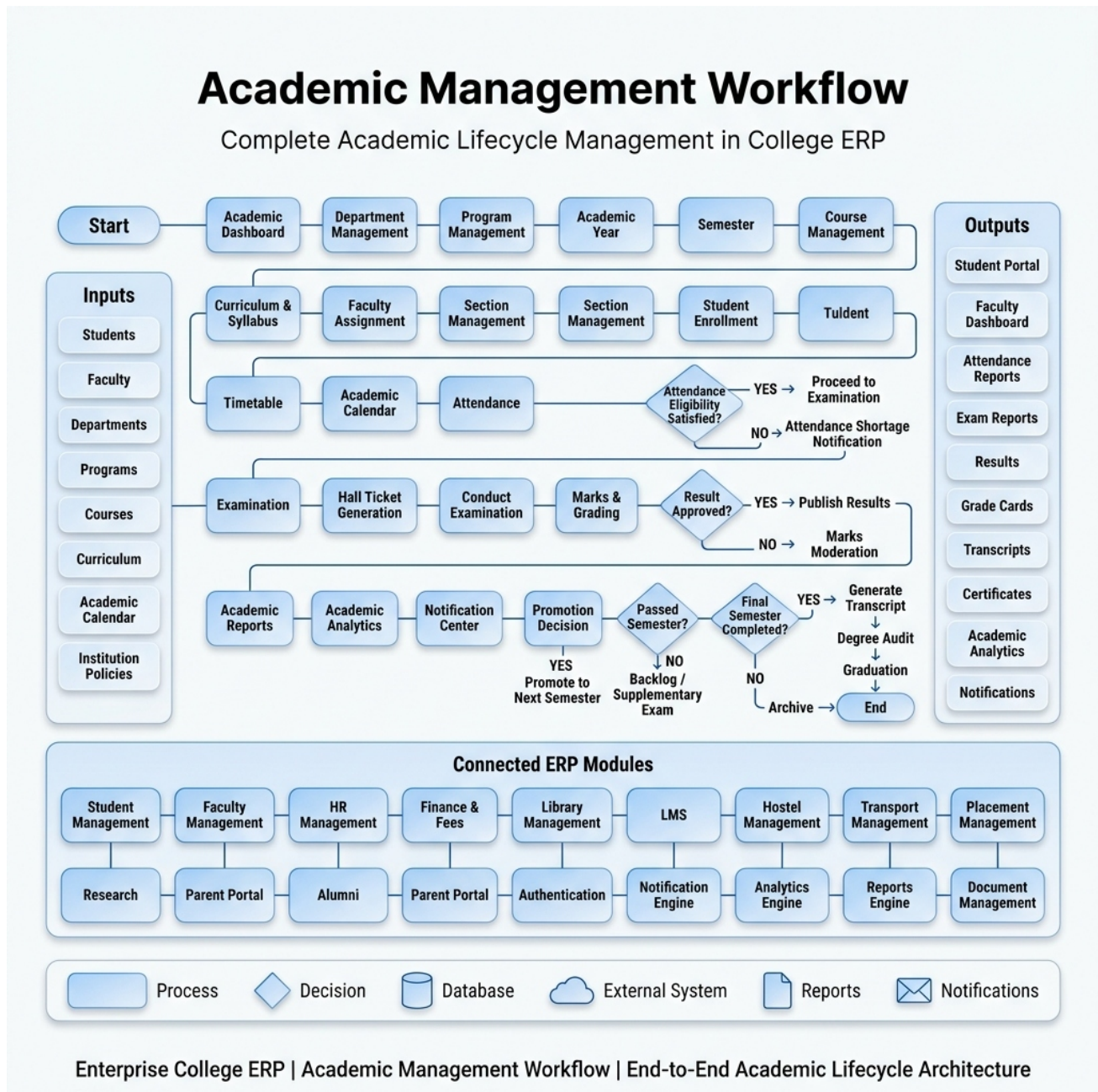


Figure 1.1: Academic Management Lifecycle & Module Connections

2. Faculty Operations & Management

The Faculty Management Workflow defines the onboarding, performance monitoring, class assignment, research tracker, and payroll updates for teaching staff. This module enables automated attendance capture, leave workflow approvals, and promotion reviews.

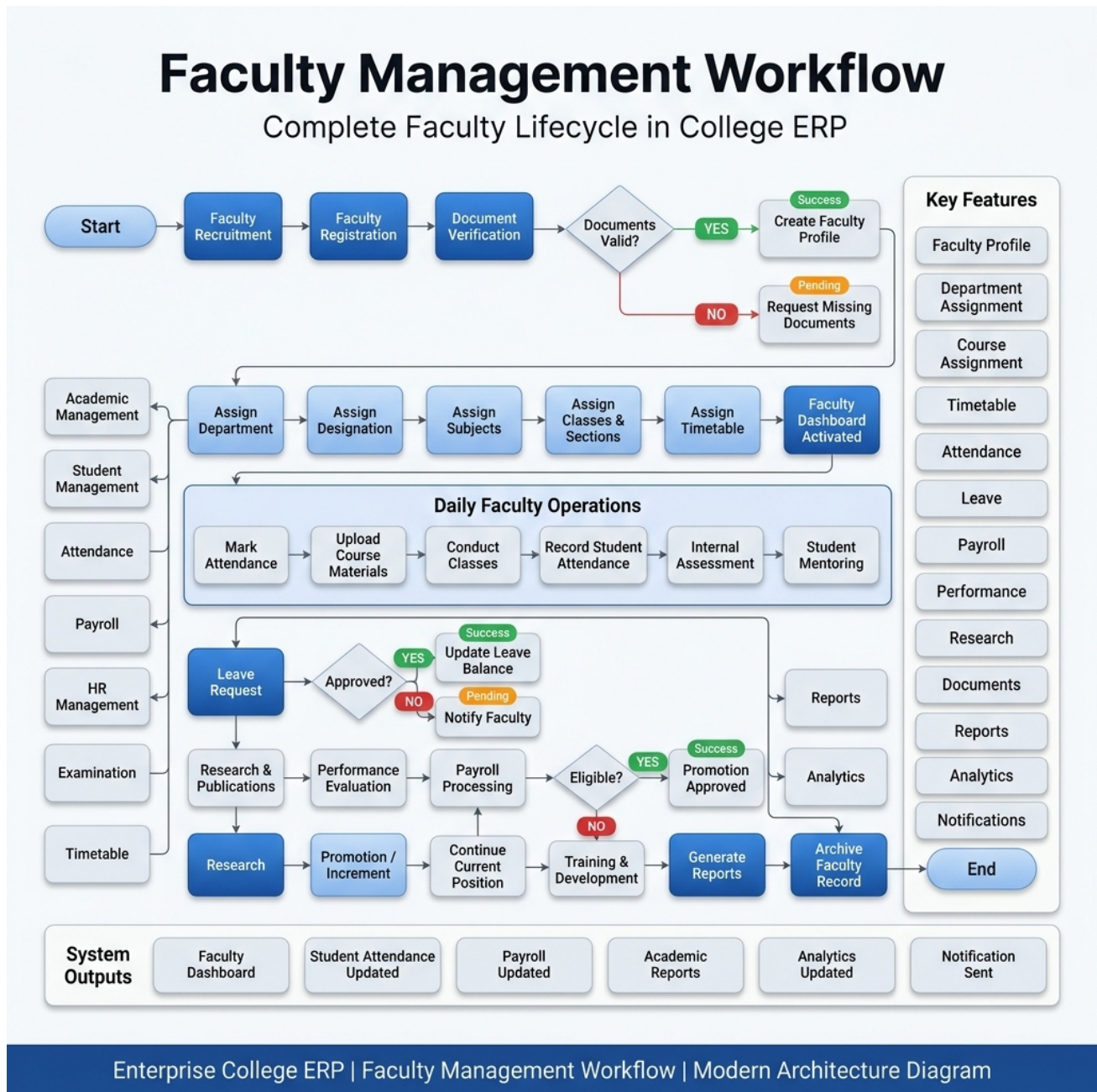


Figure 2.1: Complete Faculty Lifecycle and Operational Output Systems

3. Examination & Results Processing

The Examination & Results Processing workflow guides the creation of exam schedules, hall ticket approvals, secure internal/external mark entry by faculty members, combined marks aggregation, and final results publication on the student and parent portals.

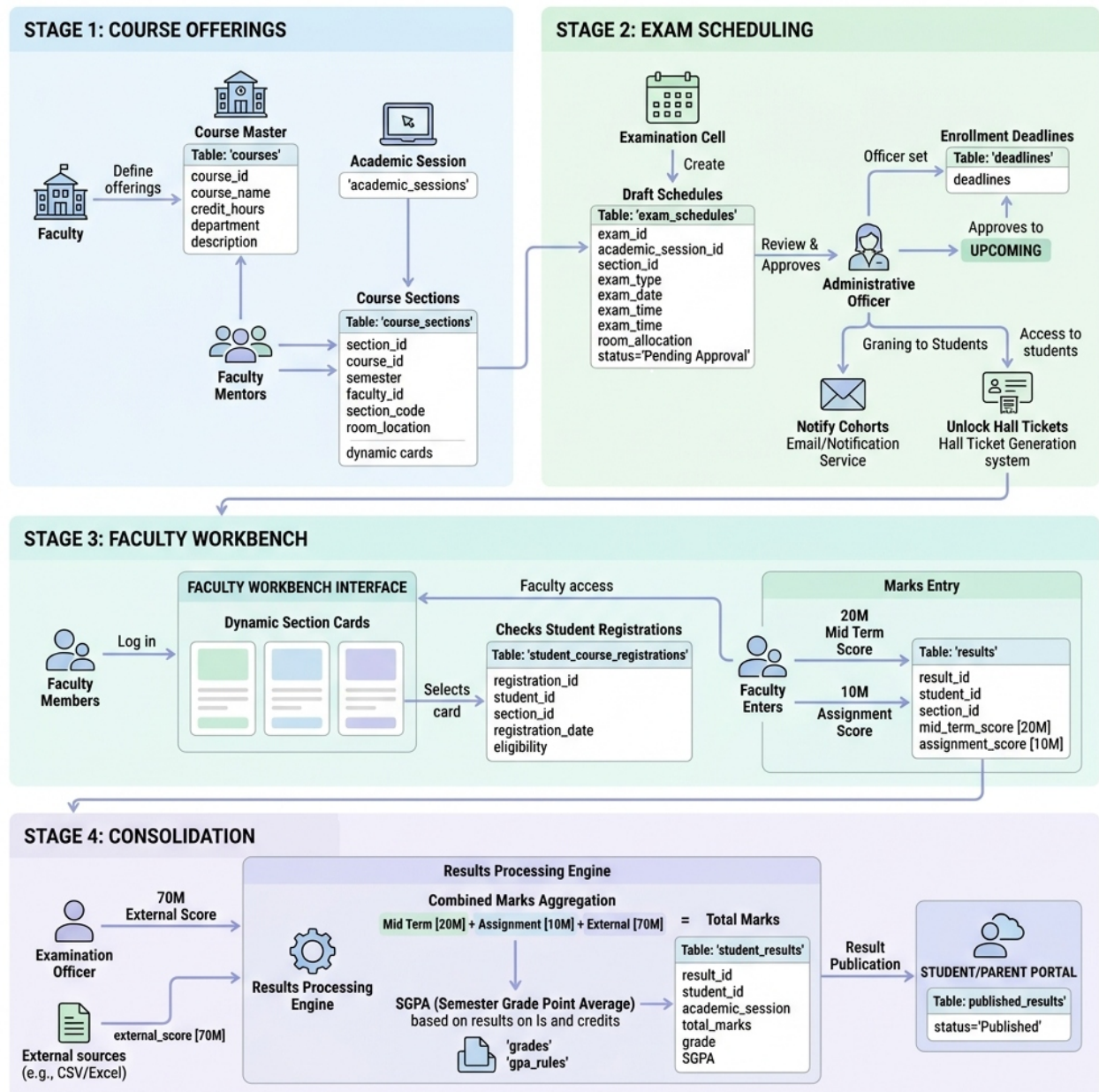


Figure 3.1: Examination Scheduling, Marks Entry and Aggregation Stages

4. Student Management & Lifecycle

The student lifecycle forms the core database structure of the ERP, mapping a student from pre-admission prospects to alumni archives.

Complete Student Lifecycle Flow

Stage 1: Admission Desk (Pre-Admission / Seat Allocation / Admission)

Stage 2: Student Created & Official ID/Credentials Generated

Stage 3: Master Student Record Linked to Core Modules (Profile, Attendance, Fees, Exams, Services)

Stage 4: Semester Promotion / Branch & Section Transfers

Stage 5: Final Year Completion & Graduating Clearance Checks

Stage 6: Alumni Registry Auto-Onboarding & Directory Archival

Student Management Module Structure

- **Admission Desk:** Pre-admission screening, Seat Allocation, Document Verification, and ID generation.
 - **Student Profile:** Personal, parent, and medical details, doc repository, and history records.
 - **Attendance Tracker:** Daily check-ins, monthly compliance sheets, and leave approvals.
 - **Examination Portal:** Internal schedules, backlog sheets, hall tickets, and grade cards.
 - **Fees & Finance:** Scholarship matching (JVD support), fee templates, online dues, and receipts.
 - **Campus Services:** Library memberships, Hostel allotments, and Transport/bus route mappings.
 - **Certificates Desk:** Automated generation of Study, Bonafide, Conduct, and Transfer Certificates.
-

4.1 Student Management Complete Workflow

This diagram details the comprehensive, end-to-end student lifecycle, illustrating the stages from initial Admission Desk registrations to final Alumni and Archive mappings.

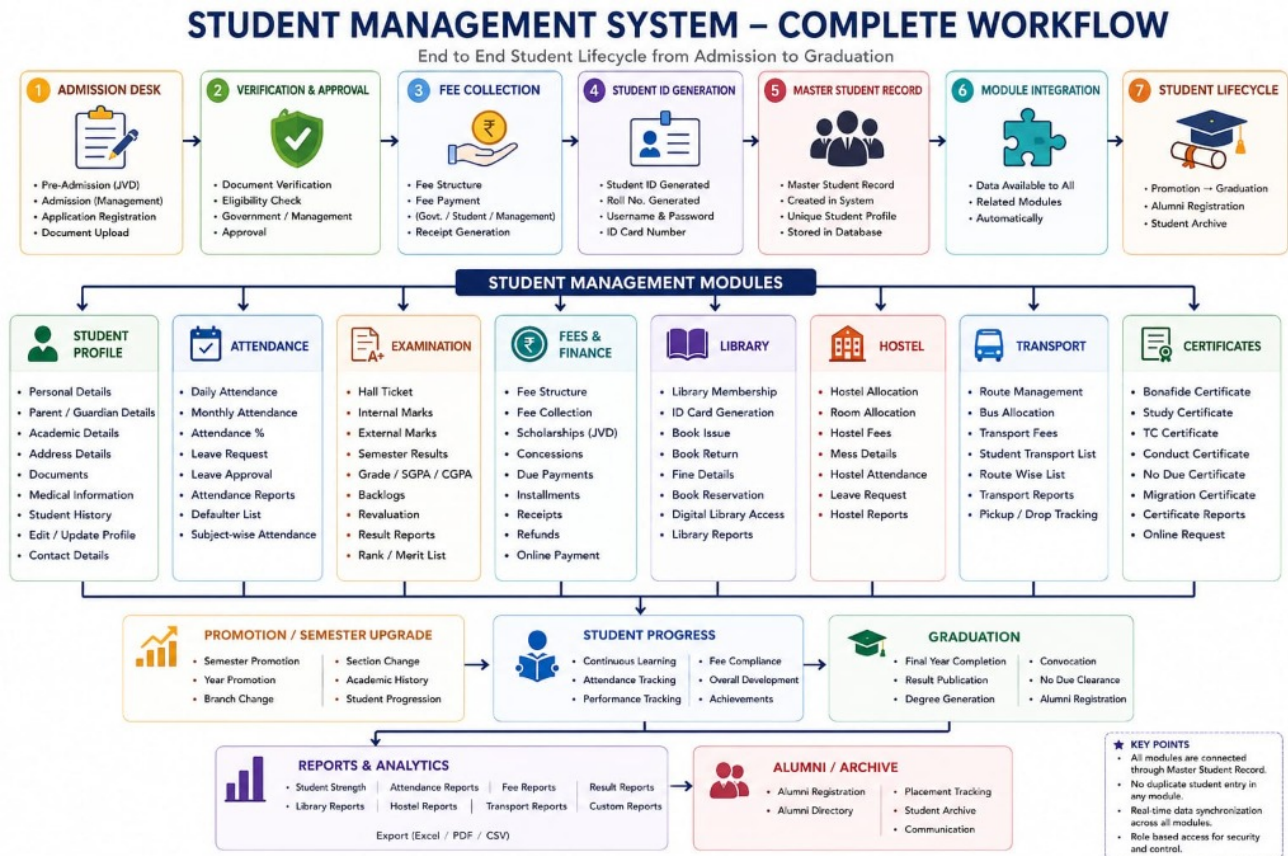


Figure 4.1: End-to-End Student Lifecycle Workflow and Core Module Integrations

5. Access Control & Admin/Super-Admin Architecture

EduSuite Pro enforces enterprise-grade security and module control, utilizing Role-Based Access Control (RBAC), responsibility override flags, dynamic feature-flag licensing, and configurable multi-step approval workflows.

Permission Evaluation Formula

Granular permissions are evaluated dynamically in real-time using the following central evaluation equation:

$$\text{Permission} = \text{Base Role} + \text{Responsibility Flags} + \text{Module License} + \text{Action} + \text{Scope}$$

Reorganized Responsibility Flags (Staff Overrides)

Staff privileges are expanded dynamically using categorized Responsibility Flags rather than hardcoded sub-roles:

System Privileges: isSystemAdmin (Global Bypass), isSecurityAdmin, isAuditAdmin, isUserManager.

Academic Privileges: isPrincipal, isVicePrincipal, isDean, isHod (Departmental Head), isExamController, isClassAdvisor, isMentor.

Student Services: isAdmissionsOfficer, isHostelWarden, isTransportOfficer, isPlacementOfficer.

Operations: isFinanceOfficer, isHRManager, isInventoryManager, isLibraryAdmin, isIQACCoordinator.

Configurable Approval Engine

Administrative tasks requiring multi-step authorizations (e.g. Leave approvals, purchases, marks publication) use a sequential approval chain configured in settings. Administrators build step timelines by linking approval checkpoints (e.g., Faculty Advisor -> HOD -> Dean -> Principal).

SaaS Feature-Flagging Engine

SaaS module licensing is tied directly to feature flags. If a license is toggled off: (1) sidebar navigation filters the module, (2) hasPermission blocks backend/frontend routing, and (3) dependent integrations (like the AI Copilot assistant) are disabled.

5.1 Role-Based Access Control Architecture

This diagram details the core 5 login roles, staff privilege override flags, department scopes, module access matrices, and dynamic approval chains.

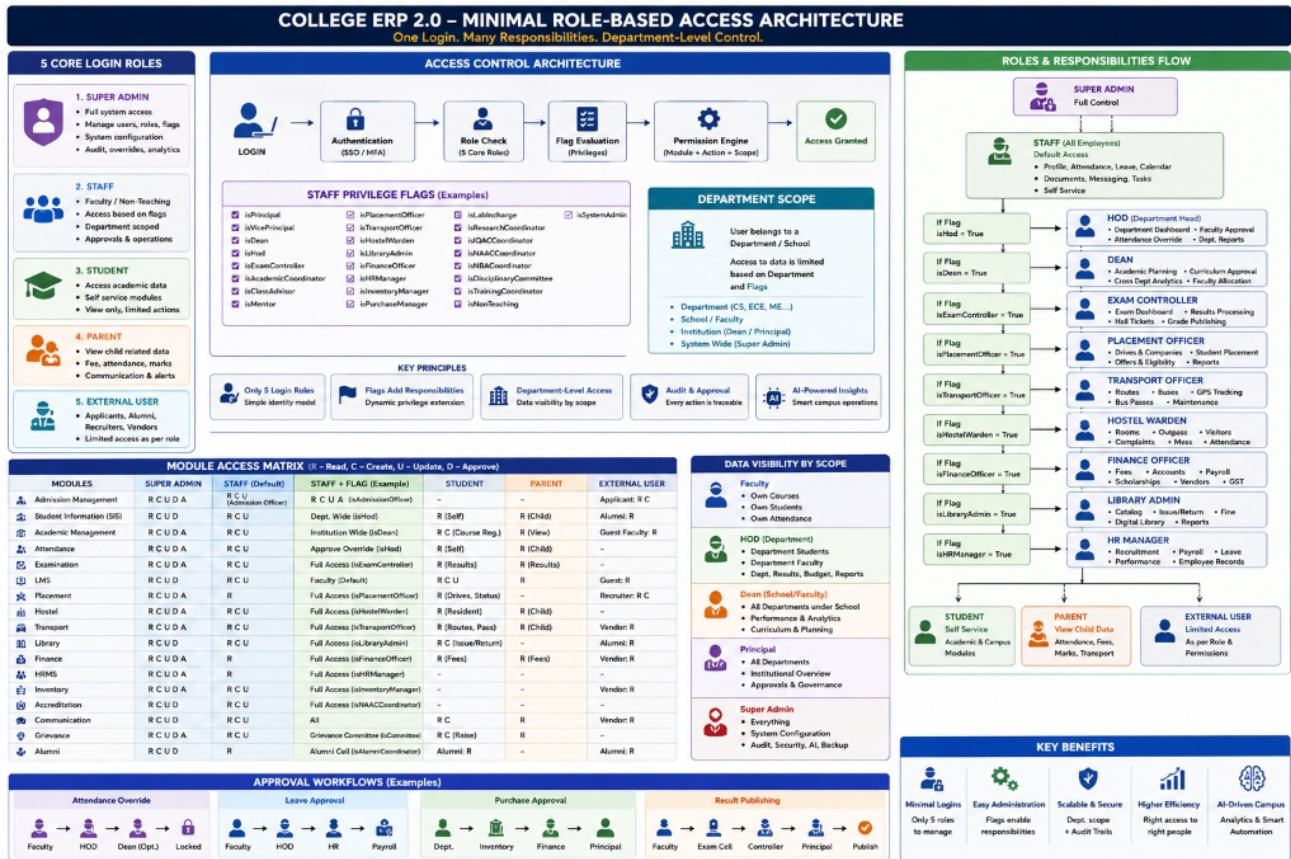


Figure 5.1: College ERP 2.0 Access Control Architecture & Responsibility Matrix

6. Training & Placement Cell Workflow

The Training and Placement (T&P) Cell module handles drive setups, auto-filtering students based on eligibility, assessment creation with dynamic IDEs, and offer letter policies.

Phase 1: Pre-Placement Preparation

- Career Declaration: Students select paths (Placement / Higher Studies / Startup).
- Profile Verification: T&P cell inspects resumes and eligibility credentials.
- Academic Validation: Automatic checks on CGPA, active backlogs, and attendance.
- Policies Setup: Configure "One Student One Job" and "Dream Offer Upgrade" conditions.

Phase 2: Company & Recruiter Onboarding

- Registration & Verification: Recruiter sign-up and security verification.
- Access Approvals: Create secure recruiter portals.
- Classification: Segment companies into Mass Recruiter, Core, Dream, and Super Dream tiers.

Phase 3: Drive Management

- Create Drive: Publish role details, salary packages, locations, and schedules.
- Auto-Filtering: System identifies eligible cohorts and triggers sign-up notifications.
- Application Reviews: Recruiter dashboard list checks.

Phase 4: Assessment Creation & Approvals

- Assessment Builder: Recruiter constructs tests containing MCQs, Coding IDE (C/C++, Java, Python), or SQL IDE environments.
- Review Queue: Placements Officer inspects, requests revisions, or approves the test for publication.

Phase 5: Assessment Operations

- Setup Test: Schedule date and send access links.
 - Test Environment: Dynamic full-screen Assessment Player with auto-save, code compilers, timers, and resume support.
-

T&P Cell Workflow (Continued)

Phase 6: Auto Evaluation Engine

- Immediate MCQ Scoring: Results saved on submission.
- Code Compiler Execution: Runs student code against pre-configured test cases (correctness, time complexity).
- SQL Query Tester: Validates student queries against database schemas and returns output scores.

Phase 7: Result Moderation

- Verification: Placement Officer audits logs and flags.
- Publication: Releases candidate shortlists to recruiters.

Phase 8: Interview Management

- Coordination: Schedule online/offline Technical and HR rounds.
- Feedback Tracking: Recruiter inputs feedback for each round.

Phase 9: Offer Management

- Upload Offers: Recruiters upload scanned/digital offer letters.
- Policy Check: System checks against student-job rules (Dream upgrades vs. blocking duplicate offers).
- Release: Verified offers released to student portals for final acceptance.

Phase 10: Analytics & Reporting Dashboard

- KPIs: Placement percentage, average package, highest package trends.
- Metrics: Grouped by Department, Companies, MCQ performance, and Interview shortlists.
- AI Insights: Predictive suggestions on cohort performance and recruitment success rates.

Phase 11: Alumni Transition

- Graduating: Auto-converts placed candidates to active Alumni status.
 - Network: Integrates student profiles with referral networks and placement drives.
-

7. Library Admin & Operations Workflow

The Library Management module automates item inventories, physical space mapping (racks/shelves), ID barcode generations, book loans, returns, and digital library resources.

Phase 1: Library Setup & Policies

- Configuration: Define loan limits (e.g., max 3 books), overdue fine rates, and academic calendars.
- Directory Mapping: Create subjects, authors, and classification hierarchies.
- Rack & Shelf Mapping: Catalog physical locations (e.g., Rack B, Shelf 4) to enable search.

Phase 2: Inventory Cataloging

- Book Intake: Input details (ISBN, title, author, edition).
- Barcoding: Auto-generate barcodes/QR codes to label book spines.

Phase 3: Member Registration

- Verification: Auto-fetch student/faculty status from master DB.
- ID Printing: Print library cards containing barcodes for fast scanner checkouts.

Phase 4: Book Issue Workflow

- Scanner Checkout: Scan student card, check limits/fines, scan book barcode.
- Transaction Log: Record transaction and compute custom due date.
- Notifications: Trigger instant issue SMS/Email receipts.

Phase 5: Book Return Workflow

- Intake: Scan book barcode.
 - Audit: Calculate overdue days and assess book condition (Good, Damaged, Lost).
 - Finalize: Update inventory stock counts and clear borrow count logs.
-

Library Operations (Continued)

Phase 6: Fine Management

- Tracking: System registers overdue charges.
- POS Checkout: Support fee payment via cash or integrated UPI.
- Receipts: Instant print and ledger synchronization.

Phase 7: Digital Library Hub

- Upload: Store PDFs, lecture notes, academic journals.
- Permissions: Match resources by department or semester rules.

Phase 8: Automated Notifications

- CRON Scheduler: Runs nightly checkups.
- Reminders: Auto-sends emails/notifications for books due tomorrow or overdue warnings.

Phase 9: Analytics & Reporting

- Operations Audit: Export reports on transaction history, catalog additions, and fine collection.
- Formats: Downloadable as PDF, Excel, and CSV.

Phase 10: Library Administration

- Management: Oversee librarian permissions and access controls.
- Insights: Real-time dashboards showing book utilization rates and peak hours.

Phase 11: Archives & Annual Audit

- Stock Verification: Run audits to crosscheck physical books against DB records.
 - Archival: Write historic circulation logs to archives.
-

